CRAFTING THE CRYPTOECONOMY · COLUMBIA UNIVERSITY

TradFi meets DeFi

Robinhood Chain and Beyond

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How big Robinhood says this is



Vlad Tenev

CEO, Robinhood · Photo: Pierce Larick

> “We’re at the beginning of a supercycle, and it’s going to take over the entire financial system.”

On tokenization · CNBC Squawk Box, 19 August 2026

> “Can we use the early success of Robinhood Chain to actually make all assets, make everything that you have in the US available to billions of people worldwide?”

On Robinhood Chain · Q2 2026 earnings call

So you’re an enterprise wanting an on-chain presence.

What should you do?

Where to build turns on three questions

1 Who sets the rules?

Ideally: You have governance sovereignty

2 What guarantees results?

Ideally: A trustless protocol

3 On what economic terms?

Ideally: Favorable and flexible

The tradeoffs are the next section of the talk. Robinhood chose to build its own Arbitrum L2.

HOW AN ENTERPRISE SHOULD CHOOSE

Where to build?

DECISION ONE

Public chain, or your own?

This decides **who sets the rules** — fees, throughput, what the chain will and will not process.

DECISION TWO

Layer 1, or Layer 2?

This decides **what guarantees results** — and what that guarantee costs in speed and money.

Public or own: what each side buys you

A public chain

- Credible neutrality
- Liquidity on day one
- No operating burden

Your own chain

- Control over fees and throughput
- Governance sovereignty
- You capture the revenue

It is a spectrum, not a switch

Public chain → **Consortium owned** → **Single firm owned**

THE NEUTRALITY PROBLEM

Would a competitor build on your chain?

If the answer is no, you have capped the chain at the size of your own business. Every rival venue has to be somewhere else, and the liquidity goes with them.

A consortium validator set is one way out — Tempo's approach. You give up sole control to buy the credibility that brings other firms in.

Who stays away, and who comes anyway

Competitors

Anyone who competes with you would be running on infrastructure you control: your fees, your throughput, your sequencing policy.

Deterred. They build somewhere else.

Complements

Anyone building a complement to your business wants to be where your users are, and can deploy across several chains at once.

Attracted, if you have the liquidity and activity.

An own chain draws complements, not competitors — and only once it is busy enough to be worth deploying on.

Two kinds of permissioned control

A BFT validator quorum

can ratify **any result it agrees on**. If the quorum says your shares moved, they moved. Nothing outside the quorum can contradict it.

The quorum decides outcomes.

L2 governance

can change the rules — **in public, and after a delay**, through whatever structure its L1 contract implements. It cannot certify a transaction the proof system rejects.

Programmed governance decides the rules.
The rules determine outcomes.

This is the distinction that makes an enterprise chain tolerable to the people transacting on it.

Governance sovereignty

Nobody can change the rules without your consent.

WHY ENTERPRISES WANT IT

Business strategy

The ability to execute it: change the chain when the product needs it, not when someone else's roadmap allows.

Chain economics

Control over fees, throughput and where the revenue goes.

Compliance obligations

Meeting them without depending on someone else's governance decisions.

L2 governance is code on L1

The rule-setting can be bounded too: the limits are contracts, chosen when you deploy, not promises.

Timelocks

Longer than the exit window, so anyone who dislikes a change can leave before it takes effect

Supermajorities

A threshold high enough that no single party can act alone

Representative councils

Security council seats held by parties with standing to object, not only the operator

Co-signing

Emergency changes requiring both the chain's own council and the Arbitrum Security Council

Governance and security emergencies

A timelock protects users from bad changes. It also delays good ones, like the fix for a live vulnerability.

Whether to have a security council

Without one, every fix waits out the timelock, even while an exploit is under way.

With one, there is a fast path for emergencies, and a party that can act without the delay users otherwise rely on.

How to design it

- Who holds the seats
- What threshold it needs to act
- What it may do: pause, patch, or anything
- How its actions are disclosed and reviewed

The emergency path is part of the governance, and it is where users should look first.

Layer 1 or Layer 2

	Your own L2 on Ethereum	Permissioned BFT L1	Open, staked L1
What guarantees results	Proofs settled on Ethereum — one honest party suffices	A quorum of named institutions	A quorum of stake, backed by slashing
Cost of security	A share of Ethereum's, paid through L1 data fees	Running the validators, plus their reputations	Paying stakers for capital lockup — $1B at 5% is $50M a year, borne alone
Latency	Set by your sequencer	Fast with few, nearby validators	Slower with many open validators
Exit to Ethereum	A protocol right	A bridge you must trust	A bridge you must trust

A permissioned L1 buys speed by giving up decentralization. An L2 takes speed from the sequencer and security from Ethereum — and does not have to choose.

One chain that moved away from staked consensus

Celo moved from its own staked L1 to an Ethereum L2 in 2025 — the one direct before-and-after comparison of what security costs.

AS AN L1 · LAST MONTH

$641,540

paid to validators

AS AN L2 · 30 DAYS

$1,070

in data-availability costs

Validator payouts: token issuance plus fee sharing. L2 figure: data availability only. Source: Conduit (Sep 2025), from L2BEAT data.

The Arbitrum barbell

One open public chain

Arbitrum One. Permissionless, credibly neutral, where the liquidity and the composability live.

Many enterprise chains

Dedicated, sovereign, tuned to one firm's product — and each of them sharing revenue back.

The bar between them is a single license. Same terms at both ends, no negotiation in the middle.

The Arbitrum license

10%

of net protocol revenue

Open to anyone. You may use and modify the software on any chain, on one condition: the chain shares ten percent of its net protocol revenue.

8%

ArbitrumDAO

2%

Developer Guild

Chains that settle to Arbitrum One are exempt from the share.

Why this is the right shape

Simple

One condition, stated once. No per-deal negotiation, and the same terms whoever you are.

Sustainable

The DAO's income grows with the ecosystem's actual revenue, not with token sales or grants.

Incentive-aligned

A share of profit, not revenue, so it doesn't push fees up. A royalty on revenue would.

The Arbitrum DAO earns from the licensed chain. It does not control the chain.

PRICING

Ethereum-style pricing: the price tracks a backlog

Backlog

Grows when usage runs above the target rate, drains when it runs below.

Price

Exponential in the backlog, scaled by the target rate times an adjustment window.

Effect

Sustained demand above target pushes the price up until usage falls back to the target.

$$p = p_0 \, e^{\frac{backlog}{target \cdot window}}$$

target: sustainable rate (gas per second)
window: adjustment window (seconds)

Ethereum is approximately this, with a window of about 100 seconds.

PRICING

Pricing protects the infrastructure

LONG TERM

State growth

Every account and storage slot is kept by every node, indefinitely. Unpriced, the state outgrows what nodes can store and sync.

SHORT TERM

Compute throughput

Nodes must execute blocks as fast as they arrive. Unpriced, a burst of demand overruns them and the chain falls behind.

Fees keep traffic within what the infrastructure can sustain — over months, and over seconds.

PRICING

Pricing with two constraints

Each constraint keeps its own backlog, target and window. Their terms add in the exponent.

$$p = p_0 e^{\frac{B_L}{T_L \cdot W_L} + \frac{B_S}{T_S \cdot W_S}}$$

B: backlog · *T*: target · *W*: adjustment window

L — long term
e.g. state growth

S — short term
e.g. compute throughput

On Robinhood Chain, currently

T_L = 15 Mgas/s W_L = 86,400 s T_S = 60 Mgas/s W_S = 15 s

PRICING

Beyond protection: pricing as policy

Infrastructure protection sets a **floor** on the price. Policy can choose any price at or above that floor, to serve the owner's own goals.

Price competitiveness

Keep fees low enough to win users and apps from other venues.

Revenue

Earn from the activity the chain carries.

Price stability

Keep fees predictable, so users and apps can plan around them.

The pricing policy decides how much weight each goal gets.

BACK TO ROBINHOOD

Robinhood Chain's three answers

Question	Answer	Implications
1 · Who sets the rules?	Its own chain	Its own customers and product; fees and throughput tuned to a brokerage, not shared with every other tenant
2 · What guarantees results?	Ethereum, as an L2	Settlement anyone can check or enforce, without funding a validator set forever
3 · On what economic terms?	The standard license	The same terms any chain gets, with its governance left in its own hands

Chain owner sets the rules but results are guaranteed trustlessly.

What's on the chain

Launched 1 July 2026

Uniswap AMM

On-chain venue for the tokenized equities

Morpho

Lending against those positions

USDG (+ other stablecoins)

A preferred stablecoin asset

100 ms preconfirmations

Fast enough that the venue feels like a broker, not a chain

Dynamic pricing

Fees that respond to load rather than sitting fixed

Activity

6.5M transactions/day · 238K daily active addresses · $1.13B DeFi TVL

Activity: weekday averages, Oct 19–23, 2026 (growthepie). TVL: DefiLlama, Oct 26, 2026.

Beyond Robinhood: the choice follows what you sell

	Layer 1	Layer 2
Use public chain	**Issuers and funds** Want the deepest liquidity and no operating role at all	**Fintechs and apps** Want cheap blockspace and composability, not sovereignty
Own the chain	**Venues and rails for others** Clearing and payment consortia: neutrality is the product, so a shared validator set is the point	**Brokers, banks, large platforms** Own customers and own product: control matters, and Ethereum carries the guarantee

An L2 is not right for everyone. Ondo canceled its own L1 — building trust has a cost, and sometimes it is the right cost to pay.

Three questions, three answers

WHO SETS THE RULES

Whoever owns the chain

And that is a narrower power than it sounds — it is fees, throughput and sequencing policy, not who owns what.

WHAT GUARANTEES RESULTS

Proofs, settled on Ethereum

Not the operator's good behavior, and not a validator quorum that can ratify whatever it agrees on.

ON WHAT ECONOMIC TERMS

Standard license: revenue share, full sovereignty

Ten percent of net protocol revenue, the same for everyone, with governance sovereignty left where it was.

Questions

Thank you.

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